GridIQ

Power-quality assessment

Demo Corporation · Headquarters

PROVISIONAL: the assessment is still running. Results cover the elapsed period only and are not final.

Assessment **GQ-A-0002 · 30-day assessment**	Report **revision 0 · provisional preview**
Observation period (SAST) **2026-10-22 21:00 -> 2026-11-11 21:11**	Observed **20.01 days of 30**
Site address **123 Power Lane, Johannesburg**	Measurement points **Phase1 Test Meter (incomer), Main Incomer (incomer)**

1. Scope and method

Ten-minute values of every measured phase across the observation period, screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values). Observed values, derived indicators, rule findings, possible causes and recommended verification are kept separate below.

Point	Coverage	Samples	Reporting interval	Gaps >= 30 min	Meter
Phase1 Test Meter	0.0 %	3	32 s	2	device
Main Incomer	64.6 %	3 722	300 s	1	device

Coverage = samples received × the meter's median reporting interval ÷ the observation window. Missing data is never filled or shown as zero.

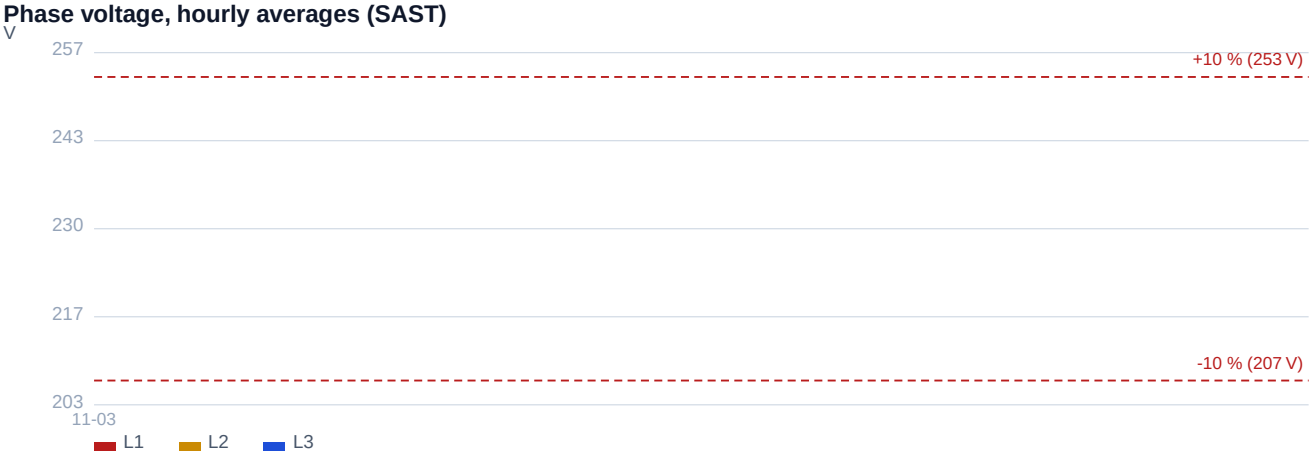
Phase1 Test Meter

Verdict type: screening only. The meter is not confirmed as an IEC 61000-4-30 Class A instrument; values are screened against NRS 048-2 thresholds, not certified.

Voltage (observed)

Phase	Min (V)	Max (V)	P5 (V)	P95 (V)	Outside ±10 % of 230 V	10-min values
A	229,9	231,5	230,5	230,5	0.0 %	1
B	229,8	229,8	229,8	229,8	0.0 %	1
C	231	231	231	231	0.0 %	1

Phase voltage, hourly averages (SAST)



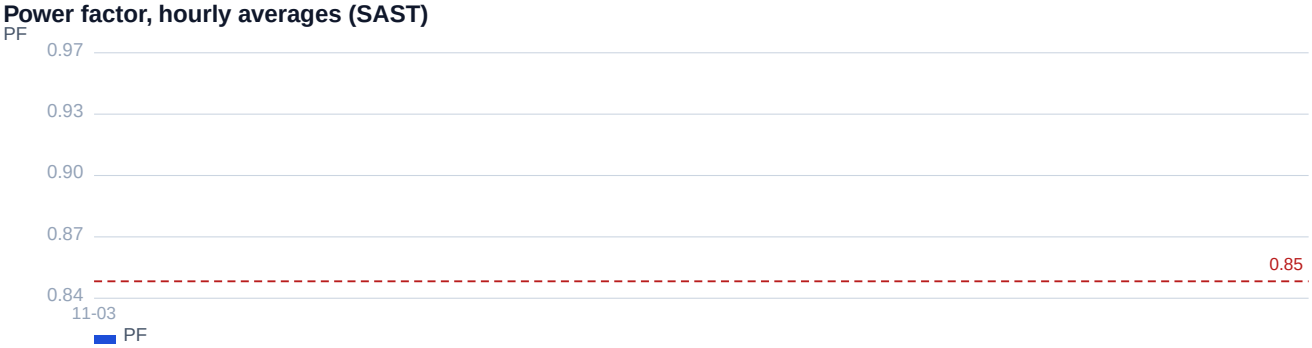
Harmonic distortion

THD of voltage was not measured at this point.

Power factor and reactive demand

Average power factor while loaded 0.957; 5th percentile 0.957; 0 % of loaded 10-minute values below 0.85. Low PF increases kVA and reactive charges; it is not a supply-quality breach.

Power factor, hourly averages (SAST)



Unbalance, frequency, interruptions and events

Voltage unbalance P95 –; current unbalance P95 – (unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Frequency 50.00 Hz–50.01 Hz; 0 10-minute value(s) outside 49–51 Hz. 0 minute(s) with voltage below 23 V.

0 excursion event(s). Source: inferred from the sampled RMS trend (pq_events), not device-recorded. device event/ waveform capture not confirmed: transient and sub-second claims are not made.

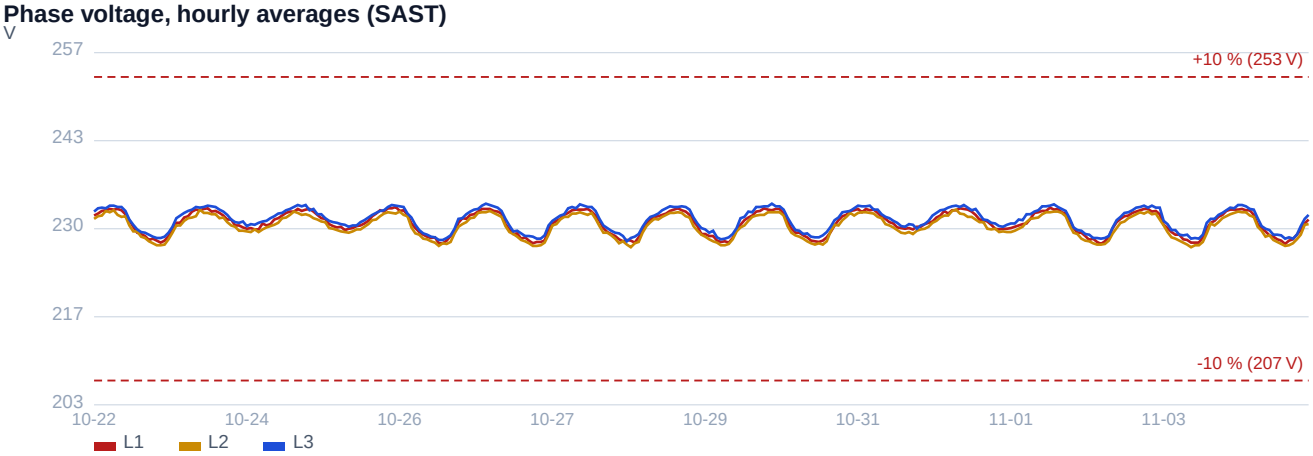
Main Incomer

Verdict type: screening only. The meter is not confirmed as an IEC 61000-4-30 Class A instrument; values are screened against NRS 048-2 thresholds, not certified.

Voltage (observed)

Phase	Min (V)	Max (V)	P5 (V)	P95 (V)	Outside ±10 % of 230 V	10-min values
A	226	234,7	228	233,3	0.0 %	1 866
B	225,7	233,9	227,5	232,8	0.0 %	1 866
C	226,6	235,6	228,5	233,8	0.0 %	1 866

Phase voltage, hourly averages (SAST)



Harmonic distortion

Phase	THD-V P95 (%)	THD-V max (%)	10-min values > 8 %	Worst hours (SAST)
A	2.69	3.08	0.0 %	13:00, 15:00, 16:00

Current THD (P95): A 12.3 %. THD-I is relative to the present current, not TDD: no IEEE 519 conclusion is drawn without the demand current and fault level.

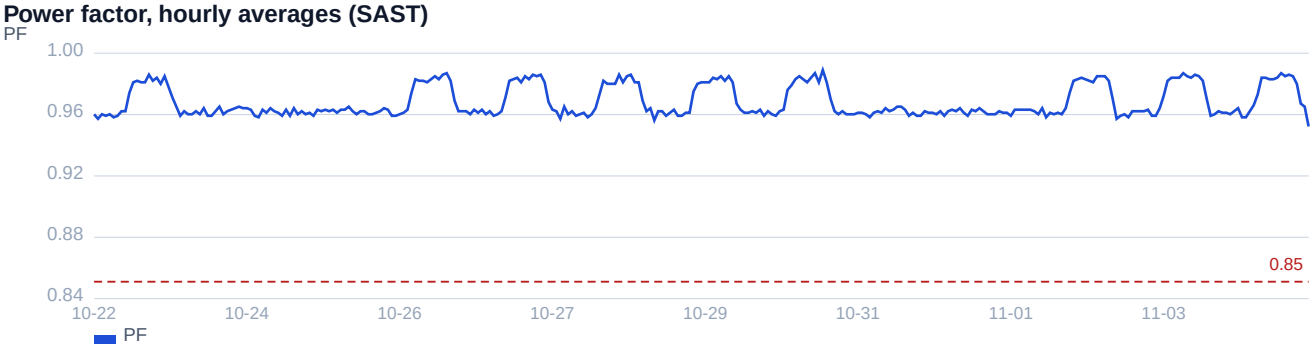
Individual orders: Individual harmonic orders were not recorded (THD only). No spectrum is shown or inferred. To identify harmonic sources, log individual orders (e.g. 3, 5, 7, 11, 13) per phase.

Voltage distortion stayed within the screening threshold in this period.

Power factor and reactive demand

Average power factor while loaded 0.968; 5th percentile 0.955; 0 % of loaded 10-minute values below 0.85. Low PF increases kVA and reactive charges; it is not a supply-quality breach.

Power factor, hourly averages (SAST)



Unbalance, frequency, interruptions and events

Voltage unbalance P95 –; current unbalance P95 – (unbalance as computed by the meter (per its configuration); not stated to be negative-sequence unbalance). Frequency 49.84 Hz–50.15 Hz; 0 10-minute value(s) outside 49–51 Hz. 0 minute(s) with voltage below 23 V.

0 excursion event(s). Source: inferred from the sampled RMS trend (pq_events), not device-recorded. device event/ waveform capture not confirmed: transient and sub-second claims are not made.

Technical appendix

Dataset schema
gridiq.assessment-dataset/1

Dataset SHA-256
86b4e86c0aedf5f1a9bcc50905be6438…

Calculation version
gridiq-calc-2026.11.1

Rule version
gridiq-rules-2026.11.1

Template version
gridiq-templates-2026.11.1

Data cutoff (SAST)
2026-11-11 21:11

Generated (SAST)
2026-11-11 21:11

Time convention
[start, end): a value at bucket B covers B <= t < B + interval

Method

- Readings are attributed to a measurement point by deployment window at the measurement time; buckets that straddle an install or removal are recomputed from the samples inside the window.
- Energy from cumulative kWh register differences per deployment; a negative step is a register reset and never summed through; power integration only where registers are absent.
- Billed demand: 30-minute minute-weighted kVA averages; instantaneous peaks are reported separately.
- Power quality: 10-minute values screened against NRS 048-2 (screening thresholds; formal compliance needs a Class A instrument, one full week and >= 95 % valid 10-minute values).
- Sign convention: import positive; the deployment sign correction is applied. Times in the data are UTC; this report shows SAST (UTC+2).

Full SHA-256 of the dataset: 86b4e86c0aedf5f1a9bcc50905be643808a12b576f4d88f9e6f9640329b8d249